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Essentials of Data Science Laboratory - 2304102L - 2025

1. Practical 1

1.1. Practice Lab Assignment

1.1.1. Calculate Momentum

1.1.2. Conditional Calculation Based on the...

1.1.3. Days between Two Dates

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1.1.1. Calculate Momentum

02:22

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:
$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).
Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

1 mass = float(input())
2 velocity = float(input())
3 momentum = mass * velocity
4 print(f"{momentum:.2f}kgm/s")

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1.1.2. Conditional Calculation Based on the Number of Digits02:51

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

$1 \leq n \leq 999$

Input Format:

The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 import math
2 n = int(input())
3 if 0 <= n <= 999:
4     if 0 <= n <= 9:
5         print(n * n)
6     elif 10 <= n <= 99:
7         print(f"{math.sqrt(n):.2f}")
8     else:
9         print(f"{n ** (1/3):.2f}")
10 else:
11     print("Invalid")
```

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1.1.3. Days between Two Dates03:58

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format *YYYY – MM – DD*.
- The second line contains the second date in the format *YYYY – MM – DD*.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

```
1 from datetime import datetime
2 date1 = input().strip()
3 date2 = input().strip()
4 date1 = datetime.strptime(date1, "%Y-%m-%d")
5 date2 = datetime.strptime(date2, "%Y-%m-%d")
6 difference = (date2 - date1).days
7 print(difference)
```

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1.1.4. Reverse a Number01:28

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 n = int(input())
2 rev = 0
3 while n > 0:
4     digit = n % 10
5     rev = rev * 10 + digit
6     n = n // 10
7 print(rev)
```

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1.1.5. Multiplication Table01:13

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

The character "x" is used in the output to represent multiplication.

Refer to the visible test-cases for more clarity.

Sample Test Cases

multipl...

1n = int(input())

2for i in range(1,11):

3 print(f"{n} x {i} = {n*i}")

4

5

6

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1.2.1. Pass or Fail

09:20

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

1 def calculate_grade(num_courses, marks):
2 v if len(marks) != num_courses:
3 . return "Error: Number of marks does not match number of courses."
4
5 v if any(mark < 40 for mark in marks):
6 . return "Fail"
7
8 . aggregate_percentage = sum(marks)/num_courses
9
10 v if aggregate_percentage > 75:
11 . grade = "Distinction"
12 v elif aggregate_percentage > 60:
13 . grade = "First Division"
14 v elif aggregate_percentage > 50:
15 . grade = "Second Division"
16 v elif aggregate_percentage >= 40:
17 . grade = "Third Division"
18 v else:
19 . grade = "Fail"
20
21 . return f"Aggregate Percentage: {aggregate_percentage:.2f}\nGrade: {grade}"
22
23
24 v try:
25 . num_courses = int(input().strip())
26 . marks = list(map(int, input().strip().split()))
27 . result = calculate_grade(num_courses, marks)
28 . print(result)
29
30 v except ValueError:
31 . print("Error: Invalid input. Please enter integers only.")

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1.2.2. Fibonacci Series

14:11

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

A single integer n representing the number of terms to generate.

Output Format:

A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

1 def fibonacci(n):

2 if n == 1:

3 return 0

4 elif n == 2:

5 return 1

6 else:

7 return fibonacci(n - 1) + fibonacci(n - 2)

8

9

10

11 n = int(input())

12 for i in range(1, n + 1):

13 print(fibonacci(i), end=" ")

14

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1.2.3. Pattern - 104:44

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.
Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 #Type Content here...
2 n = int(input())
3 for i in range(1, n + 1):
4     for j in range(i):
5         print("*",end=" ")
6     print()
```

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1.2.4. Pattern - 209:38

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

1

2

3

4

5

6

7

8

9

10

Type Content here...

Type Content here...

n = int(input())

for i in range(1,n+1):

for j in range(1,i+1):

print(j, end=" ")

print()

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